

Question Paper

Exam Date & Time: 29-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Applications of AI in Government and Business [CSE_4097]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply the steps of a typical Deep Learning Process to explain how a neural network learns to classify handwritten digits from the MNIST dataset. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Analyze how machine learning can be applied to reduce urban traffic congestion using the TranSEC model as a reference. (5)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Analyze how different types of data storage systems (SQL, NoSQL, and object storage) influence the scalability and efficiency of recommendation engines in handling user interactions and ratings. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Apply the concept of data analysis techniques in recommendation systems to determine how real-time, batch, and near-real-time methods can be used for generating relevant user recommendations in different scenarios. (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Analyze the categories used for general classification of data by recommendation systems and explain how each category contributes to generating personalized recommendations. (2)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Apply the concept of sentiment analysis to explain how businesses can improve decision-making. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Analyze the challenges posed by complex expressions and opinion characteristics in sentiment analysis and describe the elements influencing sentiment interpretation. (3)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- C) Analyze how NLP-based chatbots enhance customer service efficiency? (2)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Apply the concept of descriptive analytics to illustrate how it serves as a foundation for predictive (5)

- A) and prescriptive analytics in the evolution of AI-driven business intelligence systems.

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Apply the concept of predictive analytics to demonstrate how it is used by experts in the healthcare sector. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Analyze *any two* benefits of descriptive analytics. (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Analyze how the "Four Automation Approaches" can be utilized to redesign and improve the workflow of a government department. (5)

- A) Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Analyze how cognitive insights can be applied to large public datasets for predictive decision-making. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Apply the understanding of cognitive insight applications and list its benefits. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_4097:Applications of AI in Government and Business

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_4097:Applications of AI in Government and Business

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	6	23(46%)
4	Analysis	8	27(54%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		14	50 (100%)